

Thm. Let G be a finite group,
and let $g_1, g_2, \dots, g_n$ be representatives of the distinct conjugacy classes of G ,
not contained in the center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^n |G : C_G(g_i)|.$$

Note: $C_G(g_i) = N_G(g_i)$, and so $|G : N_G(g_i)| = C_{g_i} = \{g_j g_i g_j^{-1} \mid g_j \in G\}$.

Thm. If p is a prime number and G is group of order p^a ($a \geq 1$),
then G have a non-trivial center.

Proof. By the class equation,

$$|G| = |Z(G)| + \sum_{i=1}^k |G : C_G(g_i)|.$$

But, since $C_G(g_i)$ is a subgroup of G and $C_G(g_i) \neq G$, because $g_i \notin Z(G)$.

Therefore, by the Lagrange's Thm, p divides $|G : C_G(g_i)|$ for each i ,
so p also divides $|Z(G)|$. Thus, it cannot be trivial.

Corollary. If $|G| = p^2$ for some prime p , then G is abelian.